

Chapter 15: Language Models and the Transformer

Problem 1: Shakespeare Language Model

To satisfy the requirements of Problem 1, the prompt variable was changed to "Oh, what is our mission in the age of AI!" in order to observe how the model would predict the next text given the Shakespearean style it was trained on. Notes on how the architecture accomplished this are included in the Architecture Overview subsection below. This exercise was intended to familiarize ourselves with the model and only required a single line of code to be changed.

Problem 2: Spanish-to-English Translation

To reverse the logic in the model and convert to Spanish-to-English translation, several modifications were made to the notebook. Essentially the logic was flipped in places such as the language pair, tokenizers, dataset feature keys, model input names, and lastly the `generate_translation`. The correct results were achieved by reviewing the cell outputs after training the model. Initially, unknown tokens were being output during translation, which resulted in meticulous debugging until the desired output was achieved.

Model Architecture Overview

The model architecture for problem 1 consisted of an Embedding layer, a GRU layer, and a Dense output layer. The Embedding layer converted each character token into a dense vector representation, the GRU layer maintained a hidden state that captured sequential context across characters, and the Dense layer produced a probability distribution over the vocabulary at each step. During generation, the prompt was fed character by character to build up the hidden state, after which the model repeatedly sampled the most probable next character and fed it back in as input, producing a continuation that reflected the learned patterns of the desired writing. This will be similar for both problems.

The model architecture for problem 2 consisted of a PositionalEmbedding layer, a TransformerEncoder block, and a TransformerDecoder block. The PositionalEmbedding layer combined token embeddings with positional embeddings to preserve word order information, which is critical for translation tasks. The TransformerEncoder used multi-head self-attention to build contextual representations of the Spanish input sequence. The TransformerDecoder used both masked self-attention on the English target sequence and cross-attention over the encoder output, allowing the model to align Spanish input tokens with the corresponding English output tokens at each decoding step. A Dense output layer then produced a probability distribution over the English vocabulary to generate the final translation.